OS Lab # 05

Name: Sana Malik

Reg no: FA24-BSE-076

Semester/Section: BSE/5B

ps = **Process Status**

Shows information about **running processes**.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps
    PID TTY          TIME CMD
   2911 pts/1    00:00:00 bash
   4570 pts/1    00:00:00 ps
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

- ps → Process Status
- x → **Include processes without a controlling terminal** (background/system processes owned by you)

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps x
    PID TTY      STAT   TIME COMMAND
   1663 ?        Ss     0:00 /usr/lib/systemd/systemd --user
   1670 ?        S      0:00 (sd-pam)
   1701 ?        S<sl   0:00 /usr/bin/pipewire
   1703 ?        Ssl    0:00 /usr/bin/pipewire -c filter-chain.conf
   1705 ?        S<sl   0:00 /usr/bin/wireplumber
   1706 ?        S<sl   0:00 /usr/bin/pipewire-pulse
   1707 ?        Ss     0:00 /usr/bin/dbus-daemon --session --address=systemd: -
   1711 ?        SLsl   0:00 /usr/bin/gnome-keyring-daemon --foreground --compon
   1733 tty2     Ssl+   0:00 /usr/libexec/gdm-wayland-session env GNOME_SHELL_SE
   1742 tty2     Sl+    0:00 /usr/libexec/gnome-session-binary --session=ubuntu
   1821 ?        Ssl    0:00 /usr/libexec/xdg-document-portal
   1837 ?        Ssl    0:00 /usr/libexec/xdg-permission-store
   1873 ?        Ssl    0:00 /usr/libexec/gcr-ssh-agent --base-dir /run/user/100
```

- ps → Process Status
- -f → **Full-format listing** (more details)
- -u → Show processes for a **specific user**
- ubuntu → The user whose processes you want to see

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -fu ubuntu
UID          PID    PPID  C STIME TTY          TIME CMD
ubuntu      1663       1  0 13:26 ?        00:00:00 /usr/lib/systemd/systemd --u
ubuntu      1670    1663  0 13:26 ?        00:00:00 (sd-pam)
ubuntu      1701    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire
ubuntu      1703    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire -c filter-
ubuntu      1705    1663  0 13:26 ?        00:00:00 /usr/bin/wireplumber
ubuntu      1706    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire-pulse
ubuntu      1707    1663  0 13:26 ?        00:00:00 /usr/bin/dbus-daemon --sessi
ubuntu      1711    1663  0 13:26 ?        00:00:00 /usr/bin/gnome-keyring-daemo
ubuntu      1733    1651  0 13:26 tty2     00:00:00 /usr/libexec/gdm-wayland-ses
ubuntu      1742    1733  0 13:26 tty2     00:00:00 /usr/libexec/gnome-session-b
ubuntu      1821    1663  0 13:26 ?        00:00:00 /usr/libexec/xdg-document-po
ubuntu      1837    1663  0 13:26 ?        00:00:00 /usr/libexec/xdg-permission-
ubuntu      1873    1663  0 13:26 ?        00:00:00 /usr/libexec/gcr-ssh-agent -
```

- ps → Process Status
- -f → Full-format listing
- -p → Select a process by **PID**
- 1602 → The process ID

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -fp 1670
UID          PID    PPID  C STIME TTY          TIME CMD
ubuntu      1670    1663  0 13:26 ?        00:00:00 (sd-pam)
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

top = **Table Of Processes**

It is a real-time tool for **monitoring running processes** and system resources.

It shows:

- **CPU usage** → how much processor is being used
- **Memory usage** → RAM usage
- **PID** → process ID
- **USER** → process owner
- **CPU%** → CPU used by each process
- **MEM%** → RAM used by each process
- **COMMAND** → program/process name

```
top - 14:18:50 up 52 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 319 total,   1 running, 318 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.2 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  13117.8 total,   8791.7 free,   1852.4 used,   2799.7 buff/cache
MiB Swap:   3868.0 total,   3868.0 free,      0.0 used.  11265.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1947 ubuntu    20   0 4193820 283032 125192 S   0.7   2.1   0:18.43 gnome-s+
   4641 ubuntu    20   0   14544   5952   3744 R   0.7   0.0   0:00.02 top
      1 root      20   0   23384  14612   9752 S   0.0   0.1   0:01.11 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
```

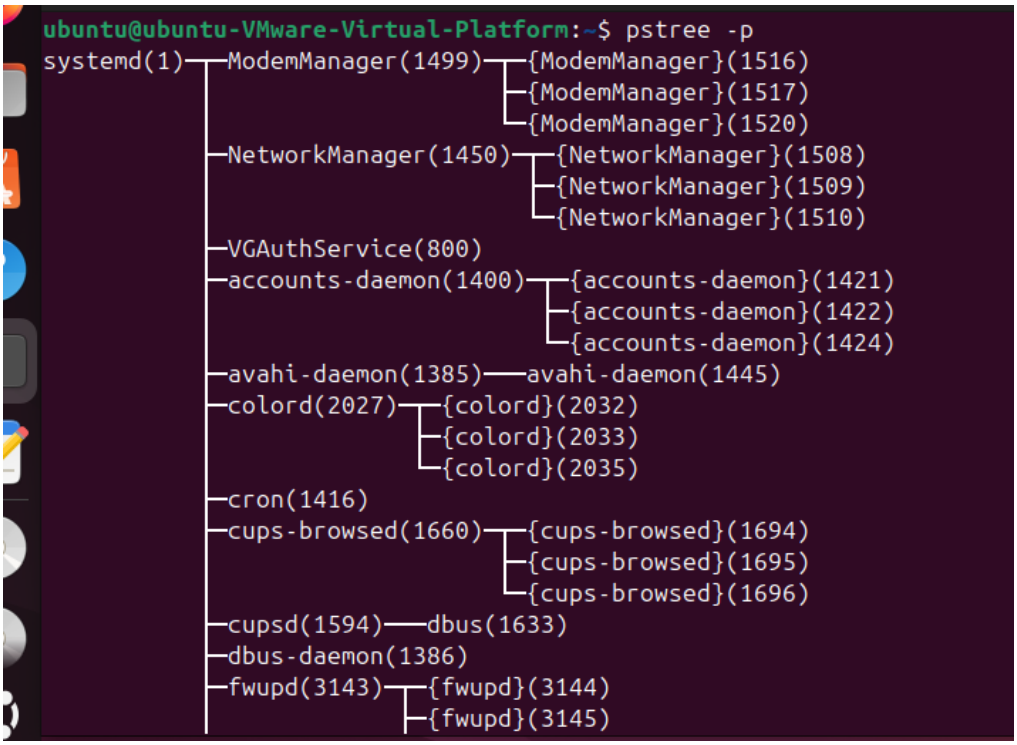
top lets us **see and manage processes in real time**.

`pstree` = **Process Tree**

It displays running processes in a **tree/parent-child structure**.



`-p` → shows the **PID** of each process.

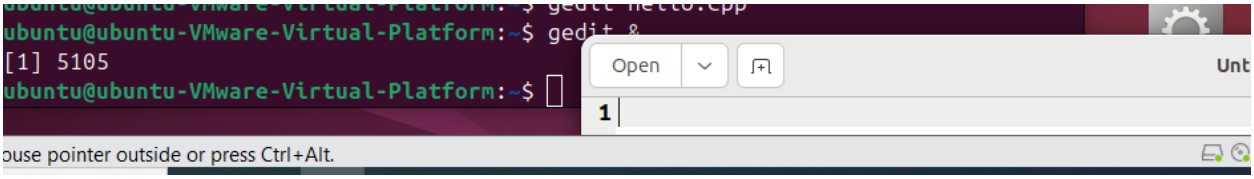


gedit **is a text editor for Linux.**

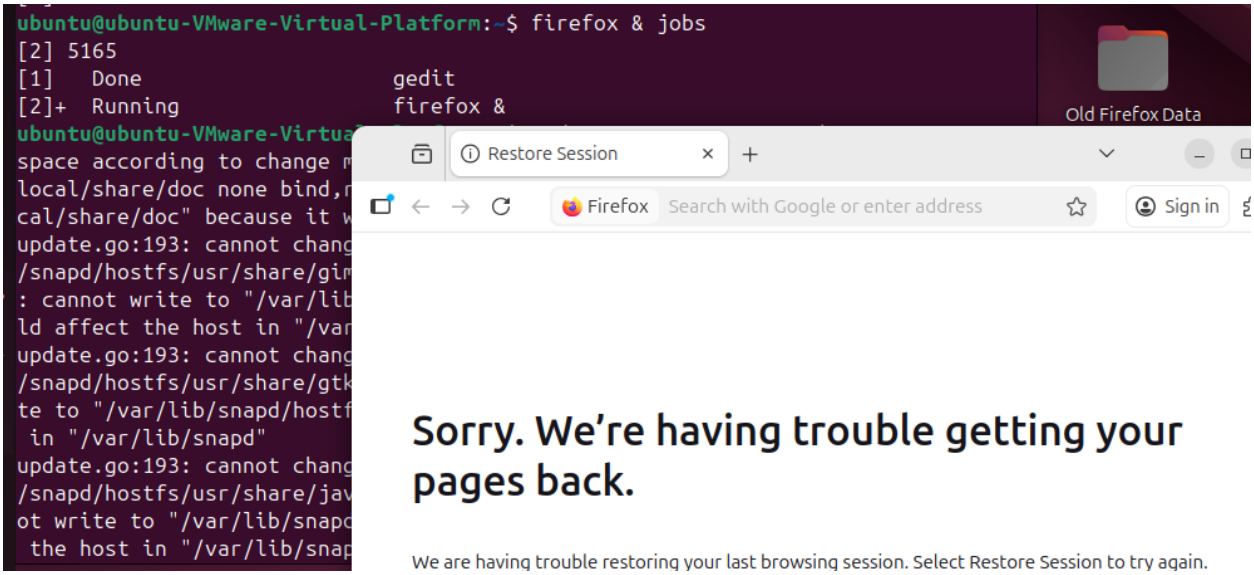
You use it to create and edit text files, including source-code files.

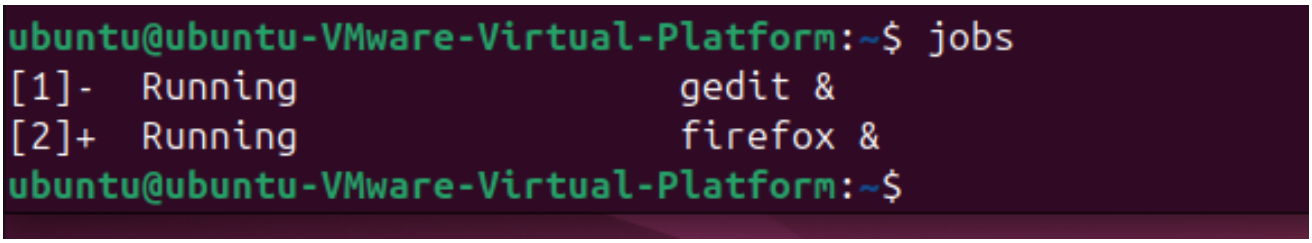
& tells Linux:

Run this command in the background.



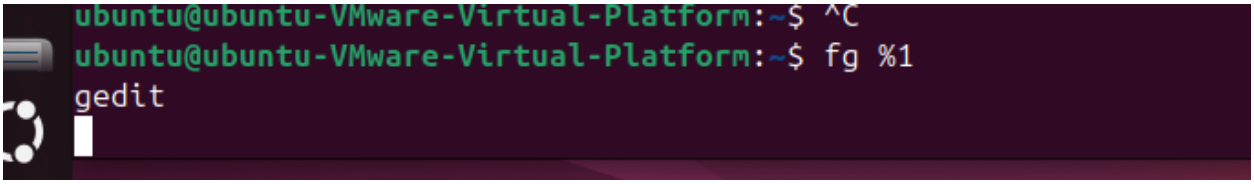
- firefox & → starts Firefox in the **background**
- jobs → shows background/stopped jobs of your **current terminal**





- fg → **foreground**
- %1 → job number **1**

→ Brings job 1 from background to the foreground.



- kill → sends a termination signal
- 6408→ PID of the process
- → Requests process **6408** to terminate.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps
    PID TTY          TIME CMD
   2911 pts/1    00:00:00 bash
   6408 pts/1    00:00:05 firefox
   6635 pts/1    00:00:00 forkserver
   6639 pts/1    00:00:00 Socket Process
   6663 pts/1    00:00:00 Privileged Cont
   6675 pts/1    00:00:00 RDD Process
   6859 pts/1    00:00:00 WebExtensions
   7178 pts/1    00:00:00 Utility Process
   7345 pts/1    00:00:00 Web Content
   7352 pts/1    00:00:00 Web Content
   7360 pts/1    00:00:00 Web Content
   7528 pts/1    00:00:00 ps
ubuntu@ubuntu-VMware-Virtual-Platform:~$ kill 6408
ubuntu@ubuntu-VMware-Virtual-Platform:~$ Exiting due to channel error.
Exiting due to channel error.
```

- kill → send a signal to a process
- -STOP → **pause/stop** the process
- 5008 → PID of the process

CONT = **Continue**

Continue running from where you stopped.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ gedit &
[1] 7567
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps
    PID TTY          TIME CMD
   2911 pts/1    00:00:00 bash
   7567 pts/1    00:00:00 gedit
   7576 pts/1    00:00:00 ps
ubuntu@ubuntu-VMware-Virtual-Platform:~$ kill -STOP 7567
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps
    PID TTY          TIME CMD
   2911 pts/1    00:00:00 bash
   7567 pts/1    00:00:00 gedit
   7582 pts/1    00:00:00 ps

[1]+  Stopped                 gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ kill -CONT 7567
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps
    PID TTY          TIME CMD
   2911 pts/1    00:00:00 bash
   7567 pts/1    00:00:00 gedit
   7588 pts/1    00:00:00 ps
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

These commands are about **process priority** — deciding which processes get more or less CPU attention.

Linux has a **nice value (NI)**:

```
Lower NI  → higher CPU priority
Higher NI → lower CPU priority
ps -lf
```

Shows processes in **long/full format**.

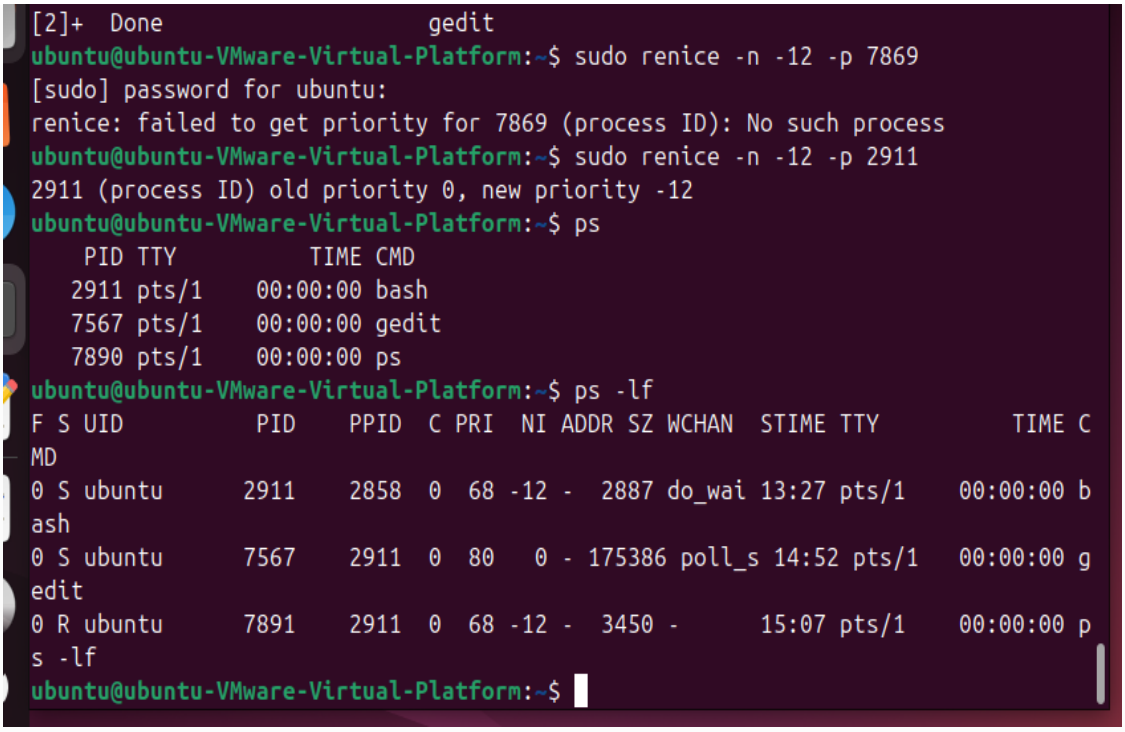
```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ gedit &
[2] 7824
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -lf
F S UID          PID    PPID  C PRI  NI ADDR SZ WCHAN  STIME TTY          TI
MD
0 S ubuntu      2911    2858  0  80   0 -  2887 do_wai 13:27 pts/1    00:00:0
ash
0 S ubuntu      7567    2911  0  80   0 - 175285 poll_s 14:52 pts/1   00:00:0
edit
0 R ubuntu      7833    2911  0  80   0 -  3450 -      15:01 pts/1    00:00:0
s -lf
[2]+  Done                    gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

- nice → start a program with a chosen nice value
- -10 → nice value **+10**
- gedit → program to start
- & → run it in the background

Since 10 is higher than normal 0, it gets **lower CPU priority**.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ nice -10 gedit &
[2] 7846
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -lf
F S UID          PID    PPID  C PRI  NI ADDR SZ WCHAN  STIME TTY          TIME C
MD
0 S ubuntu      2911    2858  0  80   0 -  2887 do_wai 13:27 pts/1    00:00:00 b
ash
0 S ubuntu      7567    2911  0  80   0 - 175318 poll_s 14:52 pts/1   00:00:00 g
edit
0 R ubuntu      7854    2911  0  80   0 -  3450 -      15:03 pts/1    00:00:00 p
s -lf
[2]+  Done                    nice -10 gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

- sudo → run with administrator privileges
- renice → **change the priority of an already-running process**
- -n -12 → set nice value to **-12**
- -p 5161 → apply it to process **PID 5161**



nice → set priority when STARTING a process

renice → change priority of an ALREADY RUNNING process

NI = -12 → higher priority

NI = 0 → normal

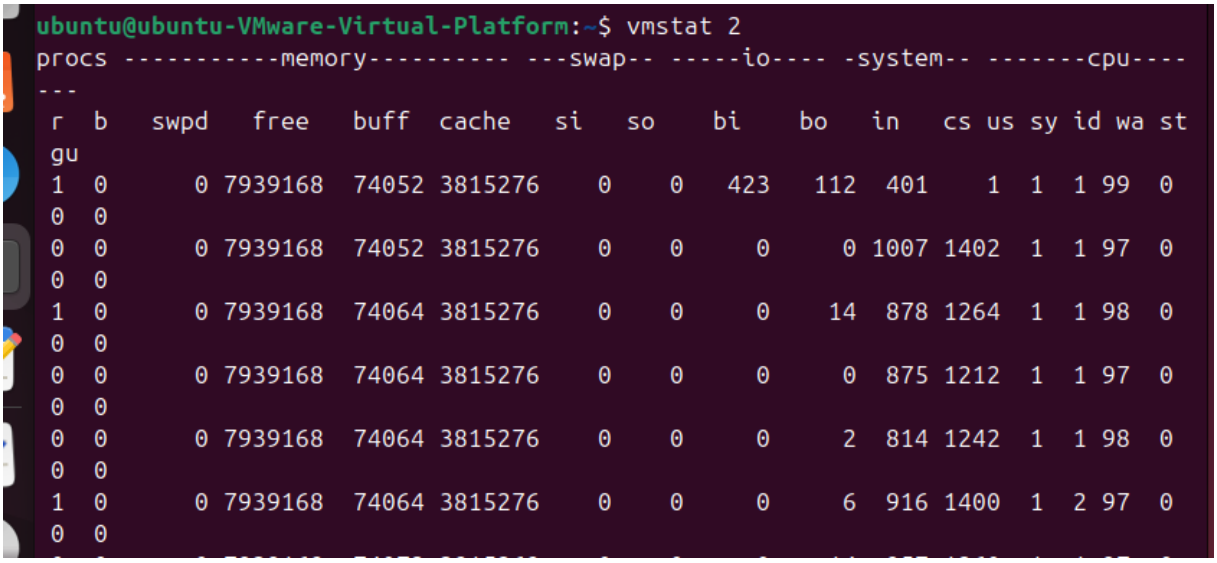
NI = +10 → lower priority

VM = Virtual Memory
stat = statistics

These are all **Linux system monitoring commands**, but each shows something a little different.

It shows statistics about:

- CPU
- RAM/memory
- processes
- swap
- disk I/O



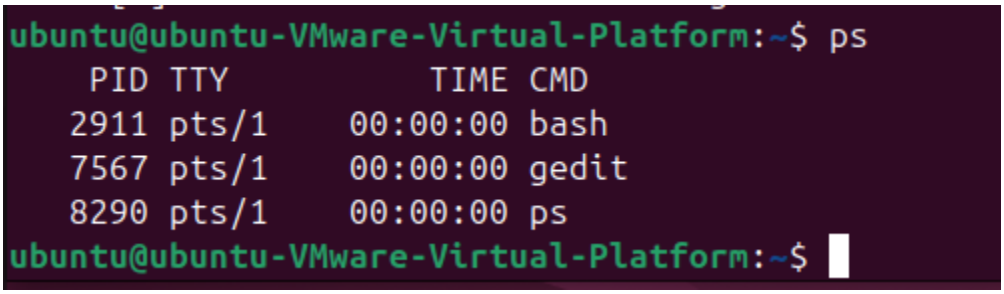
vmstat → detailed system statistics

xload → graphical load graph

tload → terminal load graph

Activity 1 — Process Monitoring in Linux

Ps ➞ Shows processes running in your **current terminal/shell**.



`ps -e` Shows **all processes** running on the system.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -e
    PID TTY          TIME CMD
      1 ?        00:00:01 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 pool_workqueue_release
      4 ?        00:00:00 kworker/R-rcu_gp
      5 ?        00:00:00 kworker/R-sync_wq
      6 ?        00:00:00 kworker/R-kvfree_rcu_reclaim
      7 ?        00:00:00 kworker/R-slub_flushwq
      8 ?        00:00:00 kworker/R-netns
     10 ?        00:00:00 kworker/0:0H-events_highpri
     12 ?        00:00:00 kworker/u512:0-ipv6_addrconf
     13 ?        00:00:00 kworker/R-mm_percpu_wq
     14 ?        00:00:00 ksoftirqd/0
     15 ?        00:00:00 rcu_preempt
     16 ?        00:00:00 rcu_exp_par_gp_kthread_worker/1
     17 ?        00:00:00 rcu_exp_gp_kthread_worker
     18 ?        00:00:00 migration/0
```

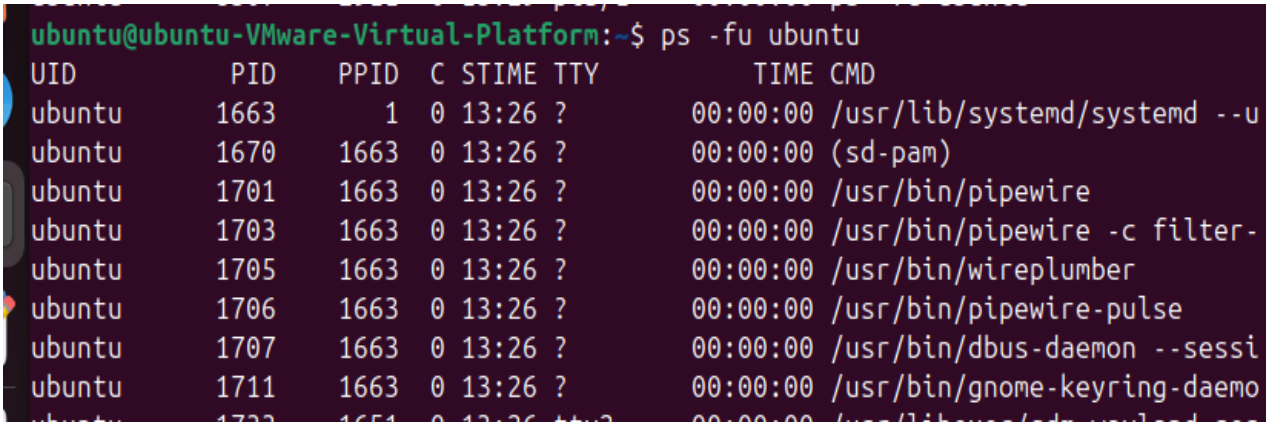
`ps -fu $USER`

-f = full format
➞ -u = user's processes
➞ $USER = your current username

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -fu $USER
UID          PID    PPID  C STIME TTY          TIME CMD
ubuntu      1663       1  0 13:26 ?        00:00:00 /usr/lib/systemd/systemd --u
ubuntu      1670    1663  0 13:26 ?        00:00:00 (sd-pam)
ubuntu      1701    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire
ubuntu      1703    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire -c filter-
ubuntu      1705    1663  0 13:26 ?        00:00:00 /usr/bin/wireplumber
ubuntu      1706    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire-pulse
ubuntu      1707    1663  0 13:26 ?        00:00:00 /usr/bin/dbus-daemon --sessi
ubuntu      1711    1663  0 13:26 ?        00:00:00 /usr/bin/gnome-keyring-daemo
ubuntu      1733    1651  0 13:26 tty2     00:00:00 /usr/libexec/gdm-wayland-ses
ubuntu      1742    1733  0 13:26 tty2     00:00:00 /usr/libexec/gnome-session-b
```

`ps -fu Ubuntu`

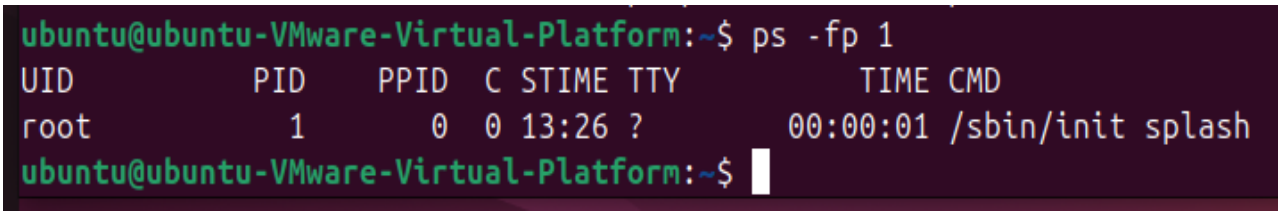
Shows all processes belonging to **ubuntu** in full format.



```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -fu ubuntu
UID          PID    PPID  C STIME TTY          TIME CMD
ubuntu      1663       1  0 13:26 ?        00:00:00 /usr/lib/systemd/systemd --u
ubuntu      1670    1663  0 13:26 ?        00:00:00 (sd-pam)
ubuntu      1701    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire
ubuntu      1703    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire -c filter-
ubuntu      1705    1663  0 13:26 ?        00:00:00 /usr/bin/wireplumber
ubuntu      1706    1663  0 13:26 ?        00:00:00 /usr/bin/pipewire-pulse
ubuntu      1707    1663  0 13:26 ?        00:00:00 /usr/bin/dbus-daemon --sessi
ubuntu      1711    1663  0 13:26 ?        00:00:00 /usr/bin/gnome-keyring-daemo
```

`ps -fp 1`

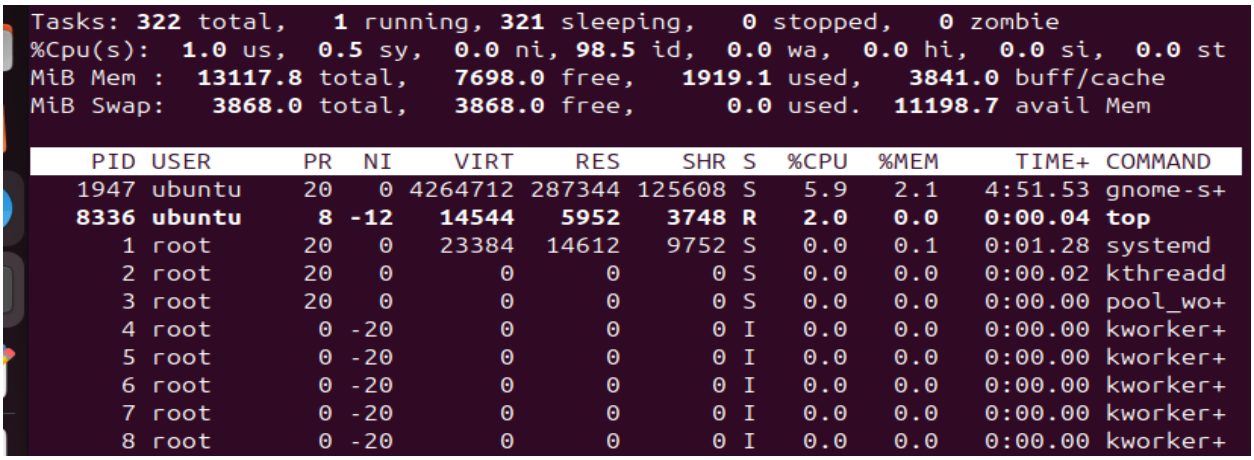
-p = specify PID
➞ 1 = process ID 1



```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -fp 1
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 13:26 ?        00:00:01 /sbin/init splash
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

`top -d 0.5`

top = dynamic process monitor
➞ -d 0.5 = refresh every **0.5 seconds**



```
Tasks: 322 total,   1 running, 321 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.0 us,  0.5 sy,  0.0 ni, 98.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  13117.8 total,   7698.0 free,   1919.1 used,   3841.0 buff/cache
MiB Swap:   3868.0 total,   3868.0 free,      0.0 used.  11198.7 avail Mem

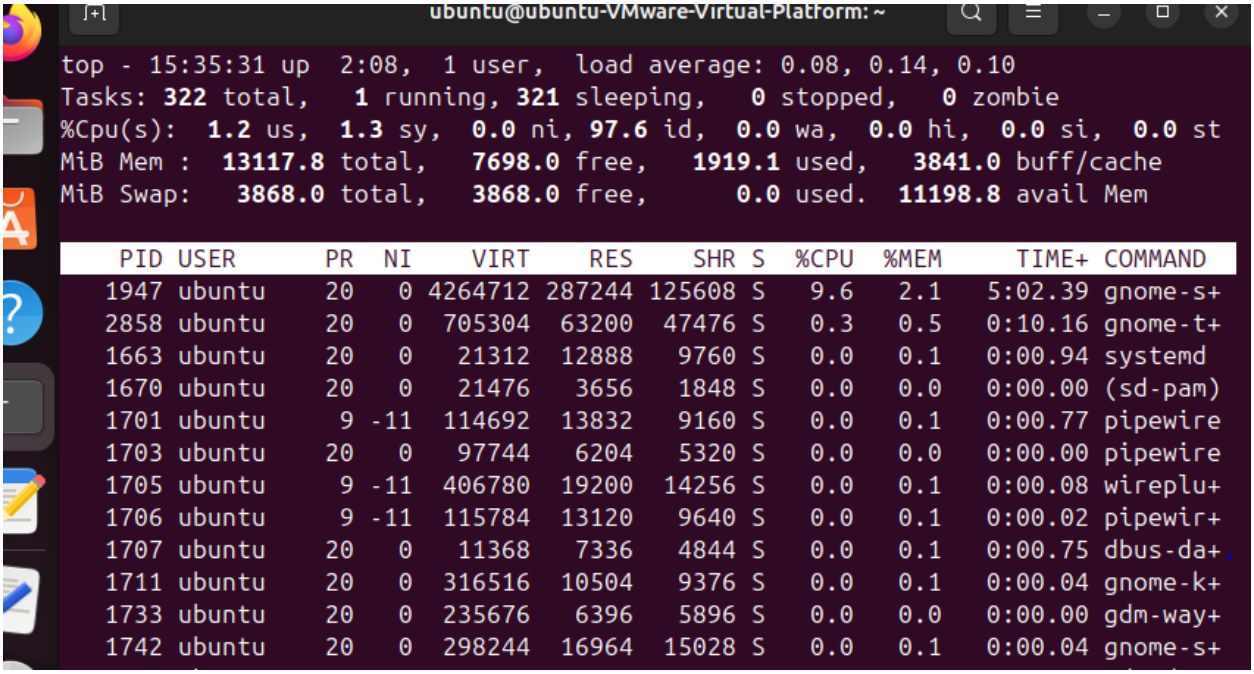
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1947 ubuntu    20   0 4264712 287344 125608 S   5.9   2.1   4:51.53 gnome-s+
   8336 ubuntu     8 -12   14544   5952   3748 R   2.0   0.0   0:00.04 top
      1 root      20   0   23384  14612   9752 S   0.0   0.1   0:01.28 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
```

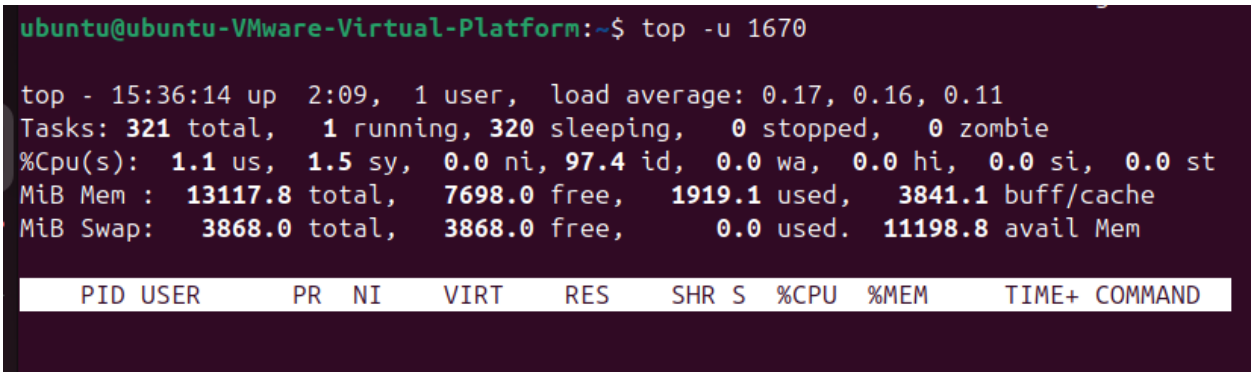
```
top -u ubuntu
```

Or by user ID:

```
top -u 999
```

Shows only processes belonging to that user.





```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ gedit &
[2] 8355
ubuntu@ubuntu-VMware-Virtual-Platform:~$ jobs
[1]-  Running                 gedit &
[2]+  Done                    gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$ fg %1
gedit
```

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ fg %1
gedit
^Z
[1]+  Stopped                 gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$ jobs
[1]+  Stopped                 gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$ bg %1
[1]+ gedit &
ubuntu@ubuntu-VMware-Virtual-Platform:~$ jobs
[1]+  Running                 gedit &
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

bg = background
→ %1 = job 1

& → start in background

jobs → show your shell's jobs

fg %1 → bring job 1 to foreground

Ctrl+Z → suspend foreground job

bg %1 → resume job 1 in background

top → dynamic process monitoring

ps → process listing

Activity 2 — Process Priority

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ nice -10 gedit &
[2] 7846
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -lf
F S UID          PID    PPID  C PRI  NI ADDR SZ WCHAN  STIME TTY          TIME C
MD
0 S ubuntu      2911    2858  0  80   0 -  2887 do_wai 13:27 pts/1    00:00:00 b
ash
0 S ubuntu      7567    2911  0  80   0 - 175318 poll_s 14:52 pts/1   00:00:00 g
edit
0 R ubuntu      7854    2911  0  80   0 -  3450 -      15:03 pts/1    00:00:00 p
s -lf
[2]+  Done                    nice -10 gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

```
[2]+  Done                    gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo renice -n -12 -p 7869
[sudo] password for ubuntu:
renice: failed to get priority for 7869 (process ID): No such process
ubuntu@ubuntu-VMware-Virtual-Platform:~$ sudo renice -n -12 -p 2911
2911 (process ID) old priority 0, new priority -12
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps
    PID TTY          TIME CMD
   2911 pts/1    00:00:00 bash
   7567 pts/1    00:00:00 gedit
   7890 pts/1    00:00:00 ps
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ps -lf
F S UID          PID    PPID  C PRI  NI ADDR SZ WCHAN  STIME TTY          TIME C
MD
0 S ubuntu      2911    2858  0  68 -12 -  2887 do_wai 13:27 pts/1    00:00:00 b
ash
0 S ubuntu      7567    2911  0  80   0 - 175386 poll_s 14:52 pts/1   00:00:00 g
edit
0 R ubuntu      7891    2911  0  68 -12 -  3450 -      15:07 pts/1    00:00:00 p
s -lf
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Activity 3: C++ Program to Find Maximum and Minimum

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ gedit maxmin.cpp
ubuntu@ubuntu-VMware-Virtual-Platform:~$ g++ maxmin.cpp -o maxmin
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ./maxmin
Maximum = 40
Minimum = 5
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[] = {10, 25, 5, 40, 15};
    int n = 5;

    int maximum = arr[0];
    int minimum = arr[0];

    for (int i = 1; i < n; i++) {
        if (arr[i] > maximum)
            maximum = arr[i];

        if (arr[i] < minimum)
            minimum = arr[i];
    }

    cout << "Maximum = " << maximum << endl;
    cout << "Minimum = " << minimum << endl;

    return 0;
}
```

Task 1 — Maximum and Minimum using Command-Line Input

```cpp
#include <iostream>
#include <cstdlib>
using namespace std;


int main(int argc, char *argv[]) {


   if (argc < 2) {
      cout << "Please enter numbers as command line arguments." << endl;
```

```
        return 1;
    }

    int maximum = atoi(argv[1]);
    int minimum = atoi(argv[1]);

    for (int i = 2; i < argc; i++) {
        int num = atoi(argv[i]);

        if (num > maximum)
            maximum = num;

        if (num < minimum)
            minimum = num;
    }

    cout << "Maximum = " << maximum << endl;
    cout << "Minimum = " << minimum << endl;

    return 0;
}
```

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ gedit maxmin.cpp
ubuntu@ubuntu-VMware-Virtual-Platform:~$ g++ maxmin.cpp -o maxmin
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ./maxmin
Maximum = 40
Minimum = 5
ubuntu@ubuntu-VMware-Virtual-Platform:~$ g++ maxmin.cpp -o maxmin
[1]+  Done                    gedit
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ./maxmin 10 25 5 40 15
Maximum = 40
Minimum = 5
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Task 2 — Palindrome Number

A **palindrome** is a number that remains the same when reversed.

121 → palindrome

1221 → palindrome

123 → not palindrome

```
#include <iostream>
using namespace std;

int main() {
   int num, original, reverse = 0, remainder;

   cout << "Enter a number: ";
   cin >> num;

   original = num;

   while (num != 0) {
      remainder = num % 10;
      reverse = reverse * 10 + remainder;
      num = num / 10;
   }

   if (original == reverse)
      cout << "Palindrome" << endl;
   else
      cout << "Not Palindrome" << endl;
```

```
    return 0;
}
```

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ gedit palindrome.cpp &
[1] 8748
ubuntu@ubuntu-VMware-Virtual-Platform:~$ g++ palindrome.cpp -o palindrome
cc1plus: fatal error: palindrome.cpp: No such file or directory
compilation terminated.
ubuntu@ubuntu-VMware-Virtual-Platform:~$ g++ palindrome.cpp -o palindrome
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ./palindrome
Enter a number: 121
Palindrome
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Task 3 — 1000 Random Numbers and Count Prime Numbers

Requirements:

- Create an array of **1000**
- Fill it with random numbers from **2 to 100**
- Check each number
- Count how many are **prime**

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

bool isPrime(int n) {
    if (n < 2)
        return false;
```

```
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0)
            return false;
    }

    return true;
}

int main() {

    int arr[1000];
    int count = 0;

    srand(time(0));

    for (int i = 0; i < 1000; i++) {
        arr[i] = rand() % 99 + 2;

        if (isPrime(arr[i]))
            count++;
    }

    cout << "Number of prime numbers = " << count << endl;

    return 0;
}
```

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ gedit prime.cpp &
[2] 8806
[1]   Done                    gedit palindrome.cpp
ubuntu@ubuntu-VMware-Virtual-Platform:~$ g++ prime.cpp -o prime
ubuntu@ubuntu-VMware-Virtual-Platform:~$ ./prime
Number of prime numbers = 244
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```
